RAKESH J

ELECTRONICS AND COMMUNICATION ENGINEERING STUDENT

CONTACT

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Nadarmedu,
Erode - 638002

EXPERTISE

- IoT
- Arduino IDE
- LTspice
- Tinkercad
- PCB Design

SKILLS

- Conflict Resolution
- Project Management
- Team Collaboration
- Continuous Learning
- Presence of Mind

PROJECTS

Laser Security System using Arduino

- Developed laser intrusion detection system.
- Implemented real-time alert generation.
- Enhanced security monitoring.

Rain Water Detector using PCB Circuit

- Designed PCB-based rain detection system.
- Automated water detection and alerting.
- Improved reliability and compactness.

LANGUAGES

- Tamil (Native)
- English

AREA OF INTERESTS

- Embedded Systems
- Internet of Things
- Digital Electronics

PROFILE

Electronics and Communication Engineering student with strong interest in Embedded Systems, Internet of Things (IoT), PCB Design, and Hardware Development. Skilled in C, Python, MATLAB, Arduino IDE, and circuit simulation tools. Experienced in designing real-time embedded solutions and IoT-based monitoring systems. Seeking opportunities to apply technical knowledge and contribute to innovative engineering projects.

EDUCATION

- **Ramco Institute of Technology, Rajapalayam**
  Bachelor of Engineering in Electronics and Communication Engineering
  2022 – 2026
  CGPA : 6.7
- **Kongu Kalvi Nilayam Matric Hr. Sec. School, Rangampalayam, Erode**
  HSC - 83.3% 2022
  SSLC - 69% 2020

CERTIFICATE COURSE

- Network Essentials in CISCO
- MATLAB Onramp in MathWorks
- Introduction to Digital Marketing in Infosys Springboard
- Printed Circuit Board Design & Analysis: From an Idea to Product – Three Day National Level Workshop conducted at Ramco Institute of Technology, Rajapalayam (Feb 26–28, 2025).

INDUSTRIAL VISIT AND INTERNSHIP

- **INDUSTRIAL VISIT**

  Kerala Electrical & Allied Engineering Co Ltd - Kochin, Kerala (June 2023)
  - Gained knowledge about working and manufacturing of transformers.

  Traco Cable Co Ltd - Kochin, Kerala (April 2024)
  - Gained knowledge about semiconductors.

- **INTERNSHIP**

  RGreen Technologies - Embedded IoT Intern
  - Worked on Embedded-IoT applications.
  - Conducted in-depth analysis of EV drive systems.
  - Integrated battery management sensors and smart monitoring.
  - Enhanced system reliability and scalability.

ACADEMIC PROJECT

IoT-Enabled Wearable Fall Detection and Health Monitoring System for Elderly People

- Designed an IoT-enabled wearable healthcare monitoring system using STM32 microcontroller.
- Integrated Accelerometer, Pulse Oximeter (SpO2), Temperature Sensor and GSR Sensor for real-time health monitoring.
- Implemented fall detection and abnormal health condition prediction using LSTM Deep Learning model.
- Enabled remote monitoring through cloud-based IoT platform.
- Generated emergency alerts for caregivers during fall events.
- Improved elderly safety through continuous health tracking, wireless communication and real-time emergency response.